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Paper Code : BSCH201 Chemistry-I (Gr-A)

UPID : 002002

Time Allotted : 3 Hours

Full Marks : 70

*The Figures in the margin indicate full marks.**Candidate are required to give their answers in their own words as far as practicable***Group-A (Very Short Answer Type Question)**

1. Answer any ten of the following :

[1 x 10 = 10]

- (I) Which type of isomerism is observed in $\text{CH}_3\text{CH}_2\text{OH}$ and CH_3OCH_3 ?
- (II) A nucleophile must possess _____.
- (III) Which type of semiconductor is formed when Germanium is doped with Aluminium?
- (IV) In UV spectroscopy, shift of λ_{max} towards shorter wavelength is called _____.
- (V) Write the expression of critical pressure.
- (VI) Give example of a reference electrode.
- (VII) Write two types of luminescence.
- (VIII) When do real gases behave as ideal gases?
- (IX) Write 3 ions which cause alkalinity of water.
- (X) Arrange NaF, NaCl, NaBr, NaI in order of increasing melting point.
- (XI) Write the criteria for a compound to be aromatic.
- (XII) How many NMR signal is obtained for isopropanol $[\text{CH}_3\text{CH}(\text{OH})\text{CH}_3]$

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. Write the differences between p-type semiconductor and n-type semiconductor. [5]
3. Explain about chromophore and auxochrome with examples. Give the range of UV spectra. [5]
4. What is MRI? State its uses. [5]
5. Explain the following two observations - the boiling point of n-pentane is higher than that of neo-pentane, H_2O is liquid while H_2S is gas. [5]
6. Starting from the expression of free energy $G = H - TS$, derive Gibbs – Helmholtz equation for constant pressure. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. (a) Show the splitting of d-orbitals in a tetrahedral field. [5]
 (b) Low spin complexes are not obtained in tetrahedral crystal field – Give reason. [3]
 (c) On the basis of band theory differentiate between conductors, semiconductors and insulators. [4]
 (d) What are anti-aromatic compounds? Give examples. [3]
8. (a) State Lambert-Beers' Law. Show that absorption is linearly proportional to the concentration of the solution. [5]
 (b) Which molecules are IR inactive? Give example. [3]
 (c) What do you mean by Bathochromic shift and Hypsochromic shift in UV spectroscopy? [4]
 (d) Which shift is observed if conjugation is increased? Give reason. [3]
9. (a) Discuss Fluorescence process with diagram. Explain its uses. [6]
 (b) Which electronic transitions are UV active for formaldehyde? Comment on their intensities of absorption. [3]
 (c) Which atoms are nmr inactive and why? [3]
 (d) What is chemical shift of proton? [3]
10. (a) Write van der Waal equation mentioning the terms involved. Show the form of van der Waal equation at high pressure and at low pressure. [4]

- (b) What is Boyle temperature. Show the relation of Boyle temperature with van der Waal's constants. [3]
- (c) What is compressibility factor? What is its value for ideal gas? [3]
- (d) Can we liquify a gas by increasing pressure alone? why? [3]
- (e) What is van der Waal forces? [2]
11. (a) With the help of a diagram, show the different electronic transitions between the molecular orbitals and comment on their energy differences. [5]
- (b) How can you differentiate 1,3-pentadiene and 1, 4-pentadiene by UV spectroscopy? [5]
- (c) A heteronuclear molecule of reduced mass 1.63×10^{-24} gm absorbs at 2880 cm^{-1} . Calculate the force constant (k) assuming harmonic oscillator model. [5]

*** END OF PAPER ***